Fiscal Year:	FY 2013	**Task Last Updated:**	FY 08/21/2013
PI Name:	Gilroy, Simon Ph.D.		
Project Title:	Mapping gravitropic signaling and response: from ions to growth		
Division Name:	Space Biology		
Program/Discipline:	SPACE BIOLOGY		
Program/Discipline--Element/Subdiscipline:	SPACE BIOLOGY--Cellular and molecular biology		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	(1) Cell & Molecular Biology (2) Plant Biology		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	53706-1313	**Congressional District:**	2
Comments:	NOTE: PI formerly at Pennsylvania State University; moved to University of Wisconsin-Madison in 2007 (Info received 7/2009)		
Project Type:	Ground	**Solicitation / Funding Source:**	2008 Space Biology--Microbial, Plant, Cell Biology: NNH08ZTT003N
Start Date:	05/27/2009	**End Date:**	05/26/2013
No. of Post Docs:	2	**No. of PhD Degrees:**	
No. of PhD Candidates:		**No. of Master' Degrees:**	
No. of Master's Candidates:		**No. of Bachelor's Degrees:**	1
No. of Bachelor's Candidates:	1	**Monitoring Center:**	NASA KSC
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:	NOTE: End date changed to 5/26/2013 per NSSC (Ed., 6/1/2012)		
Key Personnel Changes/Previous PI:	March 2012: The postdoc who was originally hired to work on the project, Dr. Won-Gyu Choi, moved on to another project and an additional postdoc, Dr. Satoe Swai-Toyota, and Assistant Scientist, Dr. Sarah Swanson, were hired to replace Dr. Choi. An undergraduate, Amanda Miller, has also been working on this project for independent study credits.		
COI Name (Institution):	Masson, Patrick (University of Wisconsin-Madison) Spalding, Edgar (University of Wisconsin-Madison)		
Grant/Contract No.:	NNX09AK80G		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	This research seeks to define the intra-/inter-cellular mechanisms behind the generation and transmission of the gravity signal. It is widely accepted that sedimentation of starch filled amyloplasts in the columella of the root cap and endodermis of the hypocotyl act to initiate the gravitropic signal. However, although pH and Ca2+-based signals have been proposed as key elements in this plant response, the precise mechanisms translating physical force to biochemically transmissible signal is still poorly defined. Needed is a high resolution spatiotemporal map of changes in ion activities induced by the gravity signal. Such mapping will be achieved by confocal imaging of plants expressing a range of GFP-based probes targeted to different subcellular locales within the Arabidopsis seedling. In parallel, we will use machine-vision technologies to quantify the distribution of curvature along the root and hypocotyl axes during gravitropism. Curvature and ion change distribution maps will be measured in wild type and mutants known to be defective in the pathways that lead to gravity signal transduction. By mapping the detailed spatial and temporal dynamics of ion signaling to these morphometric analyses, we will be able to pinpoint where and when each component is altering tropic response. Potential for translation to future spaceflight opportunities: These studies will develop a morphometric technology highly applicable to spaceflight analysis where existing hardware can be used to collect the time-lapse imaging required for morphometric analysis. Such analysis will develop a quantitative description of the effects of the flight environment on plant development at levels of architecture ranging from branching patterns to the kinematics of growth. This will allow us to ask questions only addressable by the unique environment of Space; for example, defining the effects of mutations on gravitational response thresholds.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	This work has capitalized on the development of automated image analyses algorithms that allow for the automated extraction of quantitative parameters of growth such as growth rate, organ angle, rate of curvature, and high resolution kinematic analyses of growth rates along the surface of the responding roots. Although these are being applied to root gravitropism in Arabidopsis, they hold the potential to be applicable to a much wider set of growth responses, allowing for the rapid screening of plants for quantitative differences in their growth patterns. The work has also developed the use of calcium-sensitive green fluorescent protein-based probes for plants. These probes have very high affinity for calcium allowing for the visualization of calcium signals that were previously undetectable and has applications to a wide range of stress signaling pathways in plants.
Task Progress:	This grant covered research to map the dynamics of ionic signaling associated with gravity sensing and response. We have used this analysis to assess how well such signaling events can be linked to the control of plant development and to gain insight into the initial signaling events related to gravity perception in plants. Major goals have been to: (1) develop the imaging approaches to monitor Ca2+ signaling dynamics in wild type plants and in a range of gravitropism-related mutant backgrounds, (2) apply high-resolution growth analyses to correlate signaling events to growth control, and (3) characterize the potential for cross-talk between gravitropic signaling and other response systems. (A) Rapid and asymmetrical Ca2+ changes accompany gravity perception in the root. We have used plants transformed with the GFP-based Ca2+ imaging probe cameleon YC3.6 to image the Ca2+ changes occurring in response to a range of abiotic stresses as well as gravistimulation (gravistimulation applied by rotation of the plant through 90 degrees). This sensor allowed us to resolve changes in Ca2+ levels in plants exhibiting gravitropic signaling/response with the major Ca2+ change occurring 10 minutes after gravistimulation. The timing of these changes was significant after the initiation of gravity response and we concluded most likely linked to the auxin-dependent machinery controlling growth responses, i.e. a late event in the gravitropic response following the initial events of perception. However, Ca2+ reporters with much higher sensitivities, the ‘YC-nanos’, have become available during this work and we have generated plants expressing a range of these sensors. Using these probes, we have been able to monitor Ca2+ changes that were previously undetected, most likely due to their much lower magnitude than required to alter YC3.6 signals. These Ca2+ changes occur in the lateral root cap and in the root cap columella (site of gravity perception) within 0.5-1 min of gravistimulation. This time-frame is within the presentation time of Arabidopsis, i.e. within the timing predicted for the generation of gravity signal transduction in the root cap. Unexpectedly, Ca2+ increases are also seen in mutants such as adg1 that lack starch and so are thought to not generate a directional signal to gravity in the root cap. However, the Ca2+ changes in these mutants are not linked to gravistimulation, occurring in a random pattern at the root tip in both vertically grown and gravistimulated plants. These results suggest that the starch filled amyloplasts in the root cap may be acting to localize gravitropic Ca2+-related signals in the sensory cells of the root tip and that without this strong directional component being maintained by the dense starch, the amyloplast membranes are free to interact with receptors at random. These random interactions would then generate the non-directional Ca2+ signals seen in the starchless mutants. This data supports the classic "starch-statolith" model where the mass of the starch in amyloplasts is required to reinforce the directional component of the gravity signaling system but is not required to trigger the perception machinery per se. Morphometric and kinematic analysis of the regional growth response of the root indicates all these Ca2+ changes occur well before gravitropic growth is detectable and they occur in regions of the root tip that do not show detectable cell elongation in response to gravistimulation. These data all suggest that these Ca2+ changes are likely part of the initial signaling events of gravitropic response. (B) Mutants in a range of Ca2+-related proteins alter gravitropic response. In addition to analyzing a range of gravitropic response mutants such as adg1 we have also screened mutants in Ca2+-related proteins (e.g., Ca2+ transporters and Ca2+-binding proteins that are highly expressed in the root tip) for potential effects on gravitropic response. This has defined two Ca2+ transporters (ACA1 and CAX2) and the "touch-related" genes TCH1, TCH2, and TCH3 as showing reduced gravitropic response. Mutants in the Ca2+ transporters alter root tip Ca2+ signal dynamics, providing a likely link to their effects on growth. These transporters also play a role in anoxic response of the root, with the mutants showing resistance to anoxic challenge, providing a point of cross-talk between gravitropic signaling and root zone anoxia. A major limitation to root growth in spaceflight is thought to be root zone hypoxia/anoxia and so interactions between this signaling system and that of the gravitropic response have a potentially important influence on plant growth in microgravity. The 'touch' genes (TCH1, 2, and 3) were identified from their transcriptional upregulation in response to mechanical stimulation but a role in gravitropic response has not been reported to date. We have determined that loss-of-function

	mutants in these genes show reduced gravitropic response but we have detected no clear alteration in Ca2+ signaling dynamics in these backgrounds. These observations suggest that these genes act downstream of the Ca2+ change, consistent with a role in interpreting the Ca2+ changes. All three of these TCH genes contain a Ca2+-binding EF-hand motifs. The touch-sensitivity of these genes also suggests a role in mechanical responses, suggesting they may act as a point of cross-talk between the mechanical and gravity sensing systems of the plant. (C) Hypergravity experiments support the need for sedimented amyloplasts in the gravity signaling system. As part of our analysis of the initial gravity sensing machinery, we have been collaborating with Dr. Masatsugu Toyota and the lab of Miyo Morita (NAIST, Japan) in using a centrifuge microscope to define whether amyloplast directional movements are linked to graviresponse as well as characterizing the likely role of starch in this process. This new microscope allowed imaging of amyloplast movements while stems were experiencing up to 30xG. This analysis revealed that sedimentary movements of amyloplasts under hypergravity conditions are linearly correlated with gravitropic curvature in wild-type stems. It also showed that agravitropic mutants such as sgr2 and sgr9 have immobile amyloplasts at 1xG but their gravitropic response could be rescued by growing them at 10-30 x G. This rescuing of the mutant phenotype correlated with the G force required to force their amyloplasts to sediment. Similarly, starch deficient mutants have their graviresponse rescued by the increasing G-forces that caused sedimentation even of starchless plastids. These results lend further support to the “static” or “settled” statolith model where amyloplasts in the sensory cells must sediment to the lower face of the cell to elicit a gravitropic signal, consistent with the Ca2+ signaling responses to gravistimulation.
Bibliography Type:	Description: (Last Updated: 02/02/2026)
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